# Supplementary Information for

# Coherence of Resilience Measures for the Atlantic Meridional Overturning Circulation Across a Hierarchy of Models

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This document provides the model-specific details for the four models analysed in the main text: the 3-box salinity model (Section S1), the Boussinesq latitude-depth model (Section S2), CLIMBER-X (Section S3), and PlaSim-LSG (Section S4). Figure and table numbers without the prefix "S" refer to the main text.

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# S1 Specifics on the 3-box salinity model

## State variables and units

The 5-box model of Wood et al. (2019) describes the Atlantic overturning through the exchange of salt between a North Atlantic box ($N$), a Tropical Atlantic box ($T$), a Southern Ocean box ($S$), an Indo-Pacific box ($IP$), and a deep box ($B$) below the Atlantic boxes. Temperature is not a state variable. It enters only through the prescribed difference $T_S - T_0$ in the density-driven overturning $q$. Following Alkhayuon et al. (2019), we retain only the North and Tropical Atlantic salinities as dynamical variables. The Southern Ocean and deep-box salinities vary on much longer time scales and are held fixed, while the Indo-Pacific salinity is diagnosed from conservation of the total salt content $C$. The resulting system is a two-dimensional, piecewise-smooth ordinary differential equation.

Salinities are written in the anomaly convention $S_i = 100(S_i^{\mathrm{dim}} - S_0)$, where $S_i^{\mathrm{dim}}$ is the salinity as a mass fraction and $S_0 = 0.035$. One internal unit therefore corresponds to 10 psu, and the $\pm 2$ psu perturbation region of the main text spans $\pm 0.2$ internal units. In the figures, internal values are converted as $S_i^{\mathrm{plot}} = 35 + 10 S_i$ psu, and salinity distances are multiplied by 10 to express them in psu. The prescribed $S_S$ and $S_B$ are converted to the same convention, and the Indo-Pacific salinity is evaluated at every call of the right-hand side as

$$S_{IP} = \frac{100}{V_{IP}} \left[ C - \frac{V_N S_N + V_T S_T + V_S S_S + V_B S_B}{100} - S_0 (V_N + V_T + V_S + V_{IP} + V_B) \right]. \tag{S1}$$

## Governing equations

The model allows for an additional freshwater hosing $H$ (in Sv), distributed to the North and Tropical Atlantic as $\widetilde{F}_N = F_N + 0.1311 \times 10^6 H$ and $\widetilde{F}_T = F_T + 0.6961 \times 10^6 H$. We

set $H = 0$ throughout, so that $\widetilde{F}_N = F_N$ and $\widetilde{F}_T = F_T$. The governing equations are

$$\frac{\mathrm{d}S_N}{\mathrm{d}t} = \begin{cases} \dfrac{Y}{V_N}\left[\dfrac{(q+K_N)(S_T-S_N)}{100} - \widetilde{F}_N S_0\right], & q \geq 0, \\ \dfrac{Y}{V_N}\left[-\dfrac{q(S_B-S_N)}{100} + \dfrac{K_N(S_T-S_N)}{100} - \widetilde{F}_N S_0\right], & q < 0, \end{cases} \tag{S2}$$

$$\frac{\mathrm{d}S_T}{\mathrm{d}t} = \begin{cases} \dfrac{Y}{V_T}\Big[\dfrac{q(\gamma S_S + (1-\gamma)S_{IP} - S_T)}{100} + \dfrac{K_S(S_S-S_T)}{100} \\ \qquad + \dfrac{K_N(S_N-S_T)}{100} - \widetilde{F}_T S_0\Big], & q \geq 0, \\ \dfrac{Y}{V_T}\Big[-\dfrac{q(S_N-S_T)}{100} + \dfrac{K_S(S_S-S_T)}{100} \\ \qquad + \dfrac{K_N(S_N-S_T)}{100} - \widetilde{F}_T S_0\Big], & q < 0, \end{cases} \tag{S3}$$

$$q = \frac{\lambda\left[\alpha(T_S - T_0) + \beta(S_N - S_S)/100\right]}{1 + \lambda\alpha\mu}. \tag{S4}$$

Here $V_i$ are the box volumes, $K_N$ and $K_S$ are gyre mixing coefficients, $F_N$ and $F_T$ are the net freshwater fluxes into the North and Tropical Atlantic, $\alpha$ and $\beta$ are the thermal and haline expansion coefficients, $\lambda$ is a hydraulic constant, and $\mu$ accounts for the advection of heat by the overturning. The routing parameter $\gamma$ sets the fraction of the positive overturning return flow that enters the Tropical Atlantic from the Southern Ocean rather than from the Indo-Pacific. The time-scaling factor $Y = 3.15 \times 10^9\,\mathrm{s}$ combines the number of seconds per year with the factor 100 of the anomaly convention, so that $t$ is measured in years. The overturning $q$ is in $\mathrm{m^3\,s^{-1}}$ and is divided by $10^6$ to obtain Sverdrups. When $q$ changes sign, the direction of the advective salt transport reverses, which makes the right-hand side discontinuous at $q = 0$. The two parameter vectors are listed in Table S1.

## Continuation in $CO_2$

Wood et al. (2019) calibrated the two parameter vectors against FAMOUS simulations at pre-industrial and doubled $CO_2$. Between the two calibrations, the box volumes, freshwater fluxes, temperatures, mixing coefficients, $\lambda$, $\gamma$, $\mu$, and the total salt content all change, whereas $S_S$, $S_B$, $\alpha$, $\beta$, $S_0$, and $Y$ are identical. We move along the straight line $p(r) = p_{1\times} + r(p_{2\times} - p_{1\times})$ in the full parameter space, labelled by $C(r) = 280\,\mathrm{ppm} + 280\,\mathrm{ppm}\,r$. This label is a convenient coordinate rather than a physical calibration, in particular for

Table S1: Parameters of the 3-box model for the pre-industrial ($p_{1\times}$) and doubled-$CO_2$ ($p_{2\times}$) calibrations of Wood et al. (2019) and Alkhayuon et al. (2019). Salinities $S_S$, $S_B$, and $S_0$ are given as mass fractions.

| Symbol | $p_{1\times}$ | $p_{2\times}$ | Unit |
|---|---|---|---|
| $V_N$ | $0.3261 \times 10^{17}$ | $0.3683 \times 10^{17}$ | $\mathrm{m}^3$ |
| $V_T$ | $0.7777 \times 10^{17}$ | $0.5418 \times 10^{17}$ | $\mathrm{m}^3$ |
| $V_S$ | $0.8897 \times 10^{17}$ | $0.6097 \times 10^{17}$ | $\mathrm{m}^3$ |
| $V_{IP}$ | $2.2020 \times 10^{17}$ | $1.4860 \times 10^{17}$ | $\mathrm{m}^3$ |
| $V_B$ | $8.6490 \times 10^{17}$ | $9.9250 \times 10^{17}$ | $\mathrm{m}^3$ |
| $S_S$ | 0.034427 | 0.034427 | – |
| $S_B$ | 0.034538 | 0.034538 | – |
| $S_0$ | 0.035 | 0.035 | – |
| $F_N$ | $0.384 \times 10^6$ | $0.486 \times 10^6$ | $\mathrm{m}^3\,\mathrm{s}^{-1}$ |
| $F_T$ | $-0.723 \times 10^6$ | $-0.997 \times 10^6$ | $\mathrm{m}^3\,\mathrm{s}^{-1}$ |
| $T_S$ | 4.773 | 7.919 | °C |
| $T_0$ | 2.650 | 3.870 | °C |
| $K_N$ | $5.456 \times 10^6$ | $1.762 \times 10^6$ | $\mathrm{m}^3\,\mathrm{s}^{-1}$ |
| $K_S$ | $5.447 \times 10^6$ | $1.872 \times 10^6$ | $\mathrm{m}^3\,\mathrm{s}^{-1}$ |
| $\alpha$ | 0.12 | 0.12 | $\mathrm{kg\,m}^{-3}\,°\mathrm{C}^{-1}$ |
| $\beta$ | 790 | 790 | $\mathrm{kg\,m}^{-3}$ |
| $\lambda$ | $2.79 \times 10^7$ | $1.62 \times 10^7$ | $\mathrm{m}^6\,\mathrm{kg}^{-1}\,\mathrm{s}^{-1}$ |
| $\gamma$ | 0.39 | 0.36 | – |
| $\mu$ | $5.5 \times 10^{-8}$ | $22 \times 10^{-8}$ | $°\mathrm{C\,m}^{-3}\,\mathrm{s}$ |
| $C$ | $4.4463 \times 10^{16}$ | $4.4735 \times 10^{16}$ | $\mathrm{m}^3$ |
| $Y$ | $3.15 \times 10^9$ | $3.15 \times 10^9$ | s |

$r > 1$. The continuation runs from $r = 0$ to $r = 2.4$ in steps of $\Delta r = 0.05$ (14 ppm), with one additional point at $r = 2.225$ (903 ppm). The last regular step with an on-state is $r = 2.2$ (896 ppm). At $r = 2.25$ (910 ppm), only the reversed (off) state is found, so the resilience measures are reported from 280 through 903 ppm.

Attractors are found and continued with the global continuation framework of Attractors.jl (Datseris et al., 2023), using a recurrence-based mapper on a fixed state-space grid with $S_N \in [-1, 1]$ (101 points) and $S_T \in [-0.1, 2]$ (106 points), and a seed-and-match scheme to follow attractors from one parameter value to the next. Trajectories are integrated with the stiff Rosenbrock23 solver at absolute and relative tolerances of $10^{-7}$. The on-state is identified at $r = 0$ as the attractor with the largest $q$ and is tracked along the continuation.

## Resilience estimation

The non-local measures are not computed on the fixed attractor-search grid. Instead, at every continuation step, we centre a square of half-width 0.2 internal units ($\pm 2$ psu) on the current on-state fixed point and draw 10 000 initial conditions uniformly from it. The random seed is fixed for each step to make the results reproducible. Each initial condition is integrated until it comes within $\varepsilon = 0.005$ internal units (0.05 psu) of one of the attractors. If this does not happen within 50 000 model years, the initial condition remains unassigned. This long cut-off is needed near the end of the on-branch, where the on-state recovers very slowly (the e-folding time is about 3000 years at 903 ppm) and convergence can take up to about 18 000 years.

From this ensemble, we obtain the basin volume as the fraction of initial conditions that converge to the on-state, the mean convergence time as the average first time at which these trajectories enter the $\varepsilon$-neighbourhood of the on-state, and the minimal critical shock as the smallest Euclidean distance between the on-state and a sampled initial condition that does not converge to it. The latter is limited by the sampling density and is therefore an upper bound for the true minimal critical shock. Local resilience is the inverse of the characteristic return time $\tau = -1/\mathrm{Re}(\lambda_{\max})$, where $\lambda_{\max}$ is the eigenvalue with the largest real part of the Jacobian at the on-state fixed point, obtained by forward-mode automatic differentiation. All four measures are computed with the stability-measure routines described in Morr et al. (2026).

For Figure 2, the basins are evaluated on a regular $60 \times 60$ grid over the same $\pm 2$ psu window around the on-state, using the same 50 000-year cut-off and a slightly larger proximity threshold of 0.1 psu. The four example trajectories start at offsets of $\pm 1.6$ psu in both coordinates and are shown for the first 3000 model years.

# S2 Specifics on the Boussinesq model

## Model and forcing

We use the latitude-depth Boussinesq model of Dijkstra and Molemaker (1997), studied more recently by Soons et al. (2025) and Esclapez et al. (2026). It represents an incompressible fluid in a rectangular basin, with horizontal coordinate $x \in [0, a]$, $a = 5$, and vertical coordinate $z \in [0, 1]$ increasing from the floor to the surface. The latitude label is $180x/a - 90°$. These coordinates describe an idealised meridional section; they do not prescribe an Atlantic basin volume. Temperature $T$, salinity $S$, and vorticity $\omega$ are prognostic, while the streamfunction $\psi$ is diagnosed from vorticity. With

$$u = \partial_z \psi, \qquad w = -\partial_x \psi, \qquad \omega = -\nabla^2 \psi, \tag{S5}$$

the non-dimensional equations are

$$\partial_t T + u\partial_x T + w\partial_z T = \nabla^2 T + \frac{h(z)}{\tau_T}(T_s^\gamma - T), \tag{S6}$$

$$\partial_t S + u\partial_x S + w\partial_z S = \mathrm{Le}^{-1}\nabla^2 S + \frac{h(z)}{\tau_S} S_s^\gamma, \tag{S7}$$

$$\partial_t \omega + u\partial_x \omega + w\partial_z \omega = \mathrm{Pr}\nabla^2 \omega + \mathrm{Pr}\,\mathrm{Ra}\,\partial_x (T - S), \tag{S8}$$

$$\nabla^2 \psi = -\omega. \tag{S9}$$

The factor $h(z) = \exp[-(1 - z)/\delta]$ confines thermal restoring and salinity forcing to the surface layer. The buoyancy term represents a linear equation of state: warmer or fresher water is lighter. All walls are impermeable and stress-free ($\psi = \omega = 0$), with homogeneous Neumann conditions for $T$ and $S$. Surface exchange is represented by the interior source terms, not by an additional boundary flux. Stochastic forcing and hosing are switched off.

The unmodified temperature profile is warmest at the equator,

$$T_s(x) = \tfrac{1}{2}[\cos(2\pi x/a - \pi) + 1]. \tag{S10}$$

Table S2: Parameters and numerical settings of the Boussinesq experiment. All quantities are non-dimensional.

| Symbol | Value | Meaning |
|---|---|---|
| Pr | 1 | Prandtl number, $\nu/\kappa_T$ |
| Le | 1 | Lewis number, $\kappa_T/\kappa_S$ |
| Ra | $4 \times 10^4$ | Rayleigh number |
| $a$ | 5 | Horizontal domain length |
| $\delta$ | 0.05 | Surface penetration thickness |
| $\tau_T$, $\tau_S$ | 0.1, 1 | Thermal restoring and salinity forcing scales |
| $\beta_s$ | 0 | Hemispheric asymmetry in the experiment |
| $M$, $N$ | 20, 40 | Grid intervals ($21 \times 41$ nodes) |
| $\Delta t$ | 0.001 | Integration step |
| $d$ | 0.28449 | Fractional depth of both perturbation boxes |

We vary a single parameter $\gamma$ through the forcing path

$$T_s^\gamma(x) = (1-\gamma)T_s(x) + \gamma\overline{T_s}, \tag{S11}$$

$$S_s^\gamma(x) = (1+\gamma)\,3.5\cos(2\pi x/a - \pi) - \beta_s \sin(\pi x/a - \pi/2), \tag{S12}$$

where $\overline{T_s}$ is the arithmetic mean over the horizontal grid. Increasing $\gamma$ reduces the equator-to-pole temperature contrast and strengthens salt addition in the tropics and salt removal at high latitudes. All reference states and perturbation trajectories use $\beta_s = 0$, and the imposed profiles remain fixed throughout each integration. This is a prescribed combination of reduced thermal contrast and increased freshwater forcing, rather than a calibrated response to atmospheric $CO_2$.

For comparison with the other models, we label this path by

$$C = 280\,\text{ppm}\; 2^{\gamma/0.07}. \tag{S13}$$

The coefficient is an illustrative inference from the thermal restoring target, which represents an atmospheric temperature in the model of Soons et al. (2025), rather than an imposed sea-surface temperature. Hahn et al. (2021) report CMIP6 multimodel-mean near-surface warming of 11.5 K in the Arctic, 6.4 K in Antarctica and 4.8 K in the tropics, averaged over 31 years centred on year 100 after abrupt $CO_2$ quadrupling. To obtain a

symmetric illustrative forcing, we average the two polar responses. Retaining an assumed baseline tropical-to-polar contrast of 30 K and logarithmic scaling with $CO_2$, the fractional contrast reduction per doubling is

$$k = \frac{(11.5 + 6.4)/2 - 4.8}{2 \times 30} = 0.06917 \approx 0.07. \tag{S14}$$

This is our reporting convention, not a calibration supplied by that study. The regional averaging, transient response and assumed baseline contrast limit its physical interpretation; the same calculation from the CMIP5 means gives $k = 0.055$. With the adopted $k = 0.07$, a nominal $CO_2$ doubling reduces the imposed thermal contrast by 7% and increases the salinity-flux amplitude by 7%. The latter equality is a prescribed coupling and is not independently calibrated by the thermal evidence. The logarithmic form is motivated by radiative forcing. In this coordinate, $\gamma = 0.09$ corresponds to approximately 683 ppm and $\gamma = 0.11$ to 832 ppm.

We request $\gamma = 0, 0.01, \ldots, 0.10$, followed by 0.102, 0.104, 0.106, 0.107, 0.108, 0.1085, 0.109, 0.1095, and 0.11. Stable on-state references are verified at the first 17 values, through $\gamma = 0.1085$ (about 820 ppm). Stationary correction fails at the remaining three values, for which we report no on-state resilience measures. This failure does not establish the existence or location of a bifurcation.

## Discretisation and salt conservation

The horizontal grid is uniform, and the vertical nodes are

$$z_j = \tfrac{1}{2} + \frac{\tanh[3(j/N - \tfrac{1}{2})]}{2\tanh(3/2)}, \qquad j = 0, \ldots, N. \tag{S15}$$

Finite differences approximate spatial derivatives. Diffusion and thermal restoring are implicit, advection is explicit, and the buoyancy torque uses the updated temperature and salinity. The constant linear systems are solved using cached separable factors. Tensor-product trapezoidal weights on the actual grid define volume averages, box means and field norms. No cosine-of-latitude factor is used, since the model domain is rectangular.

After each provisional salinity step, a spatially uniform correction restores the previous salt inventory:

$$S^{n+1} = \widetilde{S}^{n+1} + \langle S^n \rangle_V - \langle \widetilde{S}^{n+1} \rangle_V. \tag{S16}$$

Here $\langle \cdot \rangle_V$ denotes the quadrature-weighted domain mean, initialised to one for salinity. This correction preserves salinity gradients. A negative non-dimensional salinity is not itself a numerical failure, since the additive origin of $S$ does not affect the dynamics; nonfinite fields and failed integrations are recorded separately.

## Equilibria and local resilience

At $\gamma = 0$, an asymmetric seed is generated with $\beta_s = -0.1$ for 20 time units and then equilibrated under the symmetric forcing. Continuation follows this northern-sinking on-state to larger $\gamma$, using time integration and, near the upper end, Newton correction. The reversed state is obtained by reflecting the on-state; its residual is checked separately, and its linear stability follows from hemispheric symmetry. A separate symmetric seed at $\gamma = 0.06$ locates a haline two-cell state, which is continued through the requested forcing values. It has polar upwelling and equatorial sinking, distinguishing it from the thermal two-cell circulation. On-state and two-cell candidates must satisfy a stationarity tolerance of $10^{-9}$ and have a converged tangent calculation with a negative leading growth rate. Time-based settling is capped at 100 units and Newton correction at 20 iterations.

The full-state derivative $J_{\Delta t}$ of the implemented timestep map is used to verify linear stability. Vorticity, salinity and temperature are the independent variables; streamfunction follows from the Poisson equation. Perturbations obey the wall conditions and have zero total salt, excluding the neutral conserved-salt mode. The leading growth rate is $\max_k \mathrm{Re}[\log(\mu_k)/\Delta t]$, with $\mu_k$ the tangent multipliers. The tangent action is checked against finite differences, and eigenvalue residuals are retained.

The local-resilience values in the main comparison instead use a separate ensemble of eight salinity perturbations and one unperturbed control per available on-state. The perturbations are $(\Delta S_N, \Delta S_T) = 0.01\,\mathrm{psu}\,(\cos\theta, \sin\theta)$ for $\theta = 0°, 45°, \ldots, 315°$, with the same complete equilibrium initialization, tapers and compensation as the grid ensemble.

Each trajectory is integrated for 5 model time units (500 model years) and sampled every 0.01 units (annually). The control mean over years 10–150 defines the reference salinity. Writing the two centred box salinities as $x_t$, we fit $M$ by minimizing $\sum_t \|x_{t+1} - Mx_t\|_2^2$ over all eight trajectories and annual pairs within years 10–150. The fitted multipliers $\nu_k$ give

$$r_{\mathrm{proj}} = -\max_k \mathrm{Re}\left[\frac{\log \nu_k}{1\,\mathrm{yr}}\right]. \tag{S17}$$

The least-squares residual and conditioning are recorded. At the highest verified forcing, $\gamma = 0.1085$, an additional continuation of the 0.01 psu ensemble to 3000 model years confirms that all eight perturbations return to the full-field on-state neighbourhood for at least the final 200 years; the endpoint distance is below $7.3\times10^{-8}$. We use 0.01 psu because 0.1 psu perturbations can escape the narrow on-state basin at high forcing. Reducing the radius to 0.005 and 0.0025 psu changes the fitted rate at $\gamma = 0.1085$ by approximately 2.6% and 3.2%, respectively; the differences are smaller at the other two tested values, $\gamma = 0$ and 0.09. Fitting-window sensitivity is larger: at $\gamma = 0.09$, changing the start from year 10 to year 25 increases the estimate from 0.0119 to $0.0317\,\mathrm{yr}^{-1}$. This projected finite-window estimate need not coincide with the slowest full-state tangent decay rate, because the two salinity means do not form a closed dynamical system.

## Salinity perturbations

The North Atlantic mask covers $x \in [3.625, 5]$ (40.5°–90°N), and the Tropical mask covers $x \in [1.625, 3.625]$ (31.5°S–40.5°N). Both contain nodes with $z \geq 1 - d$, where $d = 0.28449$. Each perturbation tapers to zero at interior edges, with cosine taper widths 0.5 horizontally and $d/3$ vertically; edges at the poles and surface remain sharp. The tapered fields $b_N, b_T$ are normalised to have weighted mean one within their respective masks. For prescribed model-unit box-mean anomalies $a_N, a_T$, we apply

$$\delta S = a_N b_N + a_T b_T + c\,\mathbf{1}_R, \qquad c = -\frac{\langle a_N b_N + a_T b_T\rangle_V}{\langle \mathbf{1}_R\rangle_V}, \tag{S18}$$

where $R$ is the complement of both masks, including deeper water. This conserves total salt while leaving the two requested box-mean changes intact. Both masks are excluded from compensation even when one perturbation amplitude is zero.

The grid contains 20 equally spaced anomalies from $-2$ to $+2$ psu in each displayed coordinate, giving 400 trajectories per available on-state and 6800 trajectories in total. An additional zero-perturbation control checks each reference and is excluded from these statistics. With $\phi_* = (\omega_*, S_*, T_*, \psi_*)$ denoting the full on-state under the same forcing, every trajectory starts from

$$\phi(0) = (\omega_*, S_* + \delta S, T_*, \psi_*). \tag{S19}$$

Only salinity changes instantaneously; the existing temperature field and circulation are retained.

## Classification and finite-amplitude resilience

Trajectories are observed for at least 30 time units and may continue to 500, with diagnostics every 0.25 units and stopping checks every 5 units. We compare each trajectory with the verified full-field equilibria using

$$D(\phi, \phi_*) = \max_{f \in \{\omega, S, T, \psi\}} \frac{\sqrt{\langle (f - f_*)^2 \rangle_V}}{s_f}. \tag{S20}$$

The scales $s_f$ are fixed at their $\gamma = 0$ on-state RMS values, with floors $(1, 0.05, 0.05, 0.1)$ in the displayed field order; the salinity scale uses $S - \langle S \rangle_V$. The same scales are used at every $\gamma$. An endpoint is assigned to a target only if $D < 0.002$ at every observation in the final two time units. All 6800 grid trajectories satisfy this criterion by time 30, with no unresolved or failed cases.

The convergence time used in the paper is the first observation in the final uninterrupted sequence inside the 0.05 psu Euclidean neighbourhood of the on-state in the two-box salinity plane. Its final-inside classification agrees with full-field certification for all 6800 trajectories; eight escaping trajectories enter this projected neighbourhood

transiently but finish outside it. Its mean is conditional on returning to the on-state; a transition to another attractor is not counted as an on-state return. Because the returning set changes with forcing, we also compare the 13 perturbations that return at every available $\gamma$. Their projected mean return time decreases overall from about 102 to 81 model years, compared with 277 to 81 years for the changing set. Full-field return times at $D < 0.002$ are retained separately: their ensemble mean decreases from about 715 to 333 years by $\gamma = 0.10$, then rises to 729 years at $\gamma = 0.1085$. First-entry and dwell times and alternative thresholds are also retained as sensitivity checks. Entry times have 0.25-unit resolution, and residence through the observed endpoint does not prove residence for all future time.

The basin fraction is the number of on-state returns divided by all 400 prescribed grid points. Unresolved or failed runs would remain in that denominator. The minimal-critical-shock estimate is the smallest $\sqrt{(\Delta S_N)^2 + (\Delta S_T)^2}$ among grid points certified to reach another stable state. It is an upper bound on the minimum escape amplitude within this perturbation family. For the verified points with $\gamma \geq 0.106$, it equals the smallest sampled nonzero radius, $2\sqrt{2}/19\,\mathrm{psu} \approx 0.149\,\mathrm{psu}$, so the apparent plateau reflects grid resolution. Both measures describe a two-dimensional section of the basin; they do not measure its full volume or its closest boundary in the complete state space.

## Readouts and reporting scales

The signed overturning diagnostic is the largest-magnitude extremum of $-\psi(x_{26}, z)$ over depth, with $x_{26} = 5(26 + 90)/180$ and linear interpolation in $x$. Equivalently, with $q_N = \max_z[-\psi(x_{26}, z)]$ and $q_S = \max_z[\psi(x_{26}, z)]$, we report $q = q_N$ when $q_N \geq q_S$ and $q = -q_S$ otherwise. This convention makes northern sinking positive and the reversed and haline two-cell reference readouts negative at this latitude. Attractor identities are determined independently from the full fields. The salinity coordinates are weighted means over the two hard masks.

We use the fixed salinity conversion $S_{\mathrm{psu}} = 10.72734565\,S + 25.45554682$. Salinity differences use the slope alone, so the $\pm 2\,\mathrm{psu}$ perturbations correspond to approximately

$\pm 0.1864$ model units. For transport we normalize the unforced on-state diagnostic, $q_0 = 3.85041863647$, to 17 Sv:

$$q_{\mathrm{Sv}} = \frac{17}{3.85041863647}\, q \approx 4.41510\, q. \tag{S21}$$

The reference amplitude is motivated by the RAPID mean of $16.9 \pm 1.2$ Sv near 26.5°N over April 2004–December 2020 (Johns et al., 2023). Assigning its rounded value to the model's $\gamma = 0$ equilibrium is a reporting convention; it does not identify modern observations with a measured preindustrial state. The same positive factor multiplies every forcing value, circulation state and streamfunction field. It gives an on-state decrease from 17.00 to 15.10 Sv over the verified range, preserving the native 11.18% decline. This amplitude normalization is not a joint dimensional calibration of the basin, diffusivity and time scale.

One model time unit is reported as 100 model years, and rates are divided by 100. In the diffusive scaling of Soons et al. (2025), time is scaled by $H^2/\kappa_T$. The nominal depth $H = 1000$ m used in the geometry illustration would therefore imply $\kappa_T \approx 3.17 \times 10^{-4}\,\mathrm{m^2\,s^{-1}}$ for this time convention. This is a conditional interpretation, not an independently established ocean diffusivity. Absolute transports, salinities, times and $CO_2$ labels should be read with these choices in mind.

Figure S1 shows the three coexisting stable streamfunction fields at $\gamma = 0.09$. Figure 3 compares $\gamma = 0$ and 0.09 (280 and 683 ppm). Its four shared anomaly pairs are $(-1.3684, -1.3684)$, $(1.1579, 1.3684)$, $(1.7895, 1.7895)$ and $(1.5789, 0.1053)$ psu. The eight trajectories are replayed from their saved initial fields, recording every integration step through time 30. These dense records reproduce the original observations and endpoints; no smoothing is applied. The ensemble return-time statistics retain their original sampling interval. Basin colours distinguish on (blue), reversed (red) and two-cell (purple) outcomes; the 0.05 psu circles indicate the projected return-time threshold, with full-field endpoint certification checked separately. The reported trends remain conditional on the forcing path, perturbation depth, compensation geometry and numerical resolution.

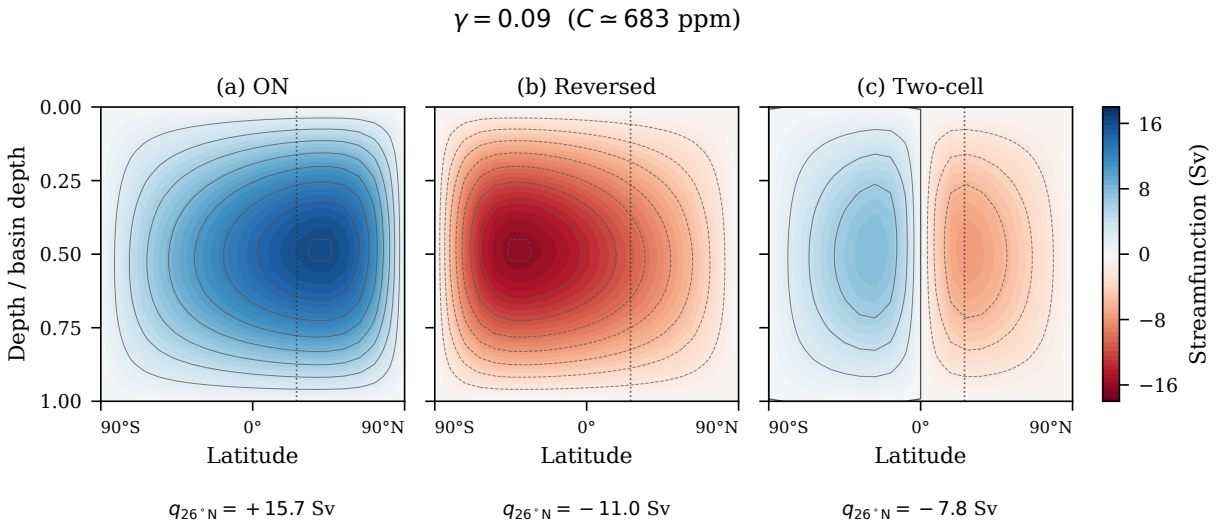



Figure S1: **Three coexisting stable Boussinesq equilibria.** The on-state, reversed state and haline two-cell state at $\gamma = 0.09$, corresponding to approximately 683 ppm on the illustrative $CO_2$ axis. Colours and contours show $-\psi$ multiplied by the same transport factor as the AMOC readout, on a shared symmetric scale. Contours are spaced by 2 Sv, with positive values solid and negative values dashed. The minus sign makes the northern-sinking on-state positive. Depth is shown as the fraction $1 - z$ of the model basin depth, with the surface at zero. The dotted section marks 26°N, where the signed AMOC strengths are 15.74, −11.04 and −7.79 Sv, respectively. All three saved equilibria satisfy the stationary and linear-stability checks described above; the transport units are an amplitude normalization, not a dimensional calibration.

# S3 Specifics on the CLIMBER-X model

## Model

CLIMBER-X v1.0 (Willeit et al., 2022) couples four components: SESAM, a 2.5-dimensional semi-empirical statistical-dynamical atmosphere; GOLDSTEIN, a three-dimensional frictional-geostrophic ocean; SISIM, a dynamic and thermodynamic sea-ice model; and PALADYN, a land surface model. All components run on a regular $5° \times 5°$ grid, and the ocean has 23 vertical layers of increasing thickness, with a top layer of 10 m. On a single node of a high-performance computer, the model simulates about 10 000 years per day. This makes ensembles of multi-millennial integrations feasible, which are out of reach for general circulation models. Atmospheric $CO_2$ is prescribed as a constant in all our runs.

## Equilibria and diagnostics

For each of the 15 $CO_2$ levels (285, 300, 315, 330, 345, 360, 385, 400, 430, 460, 490, 510, 540, 570, and 600 ppm), the model is started from the default pre-industrial restart and integrated for 20 000 years at fixed $CO_2$. The final restart serves as the reference state for all perturbation experiments at that level, and the trailing 500 years provide the reference AMOC strength. We diagnose the AMOC as the maximum of the Atlantic overturning streamfunction at 26°N.

The salinity diagnostics are written as annual means by a small modification of the model output routines. The North Atlantic box consists of all Atlantic-basin cells between 35° and 80°N, and the Tropical Atlantic box of all Atlantic-basin cells between 35°S and 35°N. Both boxes are defined by the model's basin mask, so their zonal extent follows the coastlines and is not bounded by fixed longitudes. Both include the top four ocean layers (layer centres at 5.3, 18.4, 39.4, and 78.8 m), which end at about 105 m. The box salinities are volume-weighted means and are also written for the initial state, so the time series start with the applied perturbation.

## Perturbations

Perturbations are applied directly to the salinity field of the restart file. Within each box, the anomaly is tapered with a cosine profile over two grid cells at the northern and southern box edges and over the lower two of the four layers. No tapering is applied in longitude, where the box is bounded by land. The tapered field is rescaled so that the volume-weighted mean anomaly over the box equals the prescribed value exactly. To conserve the global salt content, a uniform compensating anomaly

$$\Delta S_c = -\frac{\Delta S_{NA} V_{NA} + \Delta S_T V_T}{V_{\mathrm{glob}} - V_{NA} - V_T} \tag{S22}$$

is applied to the rest of the world ocean in the same upper 105 m, where $V_{\mathrm{glob}}$ is the volume of the upper 105 m of the global ocean and $V_{NA}$, $V_T$ are the box volumes.

In the main experiment, $\Delta S_{NA}$ and $\Delta S_T$ each take the values $-2, -1.5, \ldots, 2$ psu,

which gives 81 runs per $CO_2$ level and 1215 runs in total. Each run is integrated for 10 000 years at the $CO_2$ level of its reference state.

## Classification and resilience measures

The resilience measures are evaluated in the plane of the two box salinities, relative to the final state of the unperturbed grid run. A run is considered converged if it ends within $\varepsilon = 0.05$ psu of this reference state. Its convergence time is the time of the last entry into this $\varepsilon$-ball, after which the trajectory remains inside. Runs that do not end inside the ball are counted as not returning. They are excluded from the mean convergence time and enter the basin volume and minimal critical shock as members of a different basin. The minimal critical shock is the smallest initial distance from the reference state among these runs. Because it is resolved only to the 0.5 psu grid spacing, it is an upper bound on the true value.

At 285, 300, and all levels from 345 ppm upward, all 81 runs return. At 315 ppm, 21 runs settle on the strong state (basin volume 0.741, minimal critical shock 0.713 psu), and at 330 ppm, 46 runs do (basin volume 0.432, minimal critical shock 0.499 psu). The final states are tightly clustered. Within each cluster, they scatter by about 0.002 psu, while the strong state lies 0.068 psu (315 ppm) and 0.076 psu (330 ppm) from the reference state in North Atlantic salinity. No run ends between 0.02 and 0.05 psu from the reference state, so the classification does not depend on the exact choice of $\varepsilon$ between about 0.01 and 0.05 psu. A threshold of 0.1 psu or more, in contrast, would merge the two states. The strong-state AMOC is 0.21 Sv (315 ppm) and 0.36 Sv (330 ppm) stronger than the reference.

The strong state reached from the perturbed runs at 330 ppm has the same North Atlantic salinity (34.587 psu) as the unperturbed equilibrium at 345 ppm (34.585 psu). From 345 ppm onward, the 20 000-year equilibration itself ends on this stronger branch. The resilience measures therefore refer to the modern state at 285–330 ppm and to the strong state at 345–600 ppm. In the latter range, no perturbation within the $\pm 2$ psu grid leaves the strong-state basin, so the minimal critical shock is not defined there.

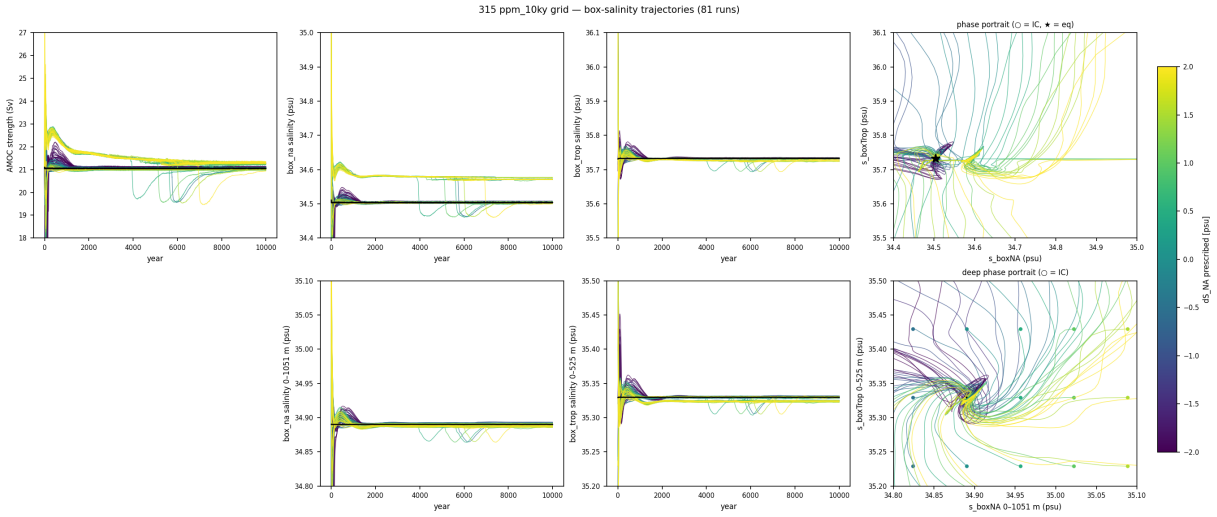



Figure S2: Trajectories of the 81 CLIMBER-X perturbation runs at 315 ppm, integrated for 10 000 years. Each run is coloured by its prescribed North Atlantic salinity perturbation $\Delta S_{NA}$; the unperturbed reference run is shown in black. The axes are zoomed onto the vicinity of the equilibria, so the initial transients and most initial conditions lie outside the plotted range. Top row: AMOC strength at 26°N, North Atlantic and Tropical Atlantic box salinities (upper 105 m) over time, and the corresponding trajectories in the box-salinity plane (star: reference equilibrium). Bottom row: the same horizontal boxes averaged over deeper layers (North Atlantic down to 1051 m, Tropical Atlantic down to 525 m), shown over time and in the corresponding salinity plane (circles: initial conditions within the plotted range). Two end states can be distinguished: the modern reference state and a strong state with about 0.07 psu higher North Atlantic salinity and a slightly stronger AMOC. Runs with positive $\Delta S_{NA}$ mostly settle on the strong state first. Several of them leave it only after 4000–7500 years, through a transient AMOC weakening of about 1.5 Sv and North Atlantic freshening, and then converge to the modern state. The 21 runs that are still on the strong state after 10 000 years are counted as not returning (Section S3).

## Local linear stability

For local resilience, nine further runs are performed at every $CO_2$ level, 135 in total. One run is unperturbed, and eight apply perturbations of 0.1 psu in directions $\theta = 0°, 45°, \ldots, 315°$, i.e. $(\Delta S_{NA}, \Delta S_T) = 0.1\,\text{psu}\,(\cos\theta, \sin\theta)$, with the same global compensation as above. Each run is integrated for 500 years. The equilibrium is taken as the mean box salinity of the unperturbed run over years 10–150, and the deviations $x_t$ of the eight perturbed runs from it are used in the same window. A $2 \times 2$ matrix $M$ is fitted by least squares to all annual pairs $(x_t, x_{t+1})$, and the generator $A = \log(M)/(1\,\text{yr})$ yields the characteristic return time $\tau = -1/\max \text{Re}(\lambda(A))$ and the local resilience $1/\tau$.

# S4 Specifics on the PlaSim-LSG model

## Model and simulations

PlaSim-LSG couples the atmospheric general circulation model of the Planet Simulator (Fraedrich et al., 2005) to the Large Scale Geostrophic ocean model (Maier-Reimer et al., 1993). The atmosphere is a spectral primitive-equation model at T21 resolution (about $5.6° \times 5.6°$) with 10 vertical levels. The ocean has an effective horizontal resolution of $3.5° \times 3.5°$ and 22 vertical layers, whose thickness increases from 50 m at the surface to 1000 m in the deep ocean. Sea ice is represented by a thermodynamic zero-layer model without ice transport. With about $10^5$ degrees of freedom, the model runs at roughly 700 model years per day on a single CPU (Börner et al., 2026). Its AMOC exhibits pronounced multicentennial variability (Mehling et al., 2023).

We use the simulations of Börner et al. (2026) at 285 and 360 ppm, available at doi:10.5281/zenodo.17053348. At both levels, the AMOC is bistable. The on-state has a mean AMOC of 14.7 Sv (285 ppm) and 15.0 Sv (360 ppm), and the off-state, in which deep convection in the subpolar North Atlantic ceases, has 1.1 and 1.7 Sv. Here, the AMOC strength is the maximum of the Atlantic overturning streamfunction between 46° and 66°N, averaged over the full equilibrium runs. The off-state at 360 ppm varies on multicentennial time scales, with 200-year means between 1.4 and 2.2 Sv.

The edge state was located with an edge-tracking algorithm (Lucarini and Bódai, 2017). Starting from a pair of states that converge to opposite attractors, nine intermediate initial conditions are generated by linear interpolation and integrated in parallel. The two neighbouring members that end up on different sides of the basin boundary define the next, ten times closer pair. Once a pair diverges by more than a set threshold in 10-year smoothed AMOC strength, it is refined again in the same way. Concatenating the segments yields a pseudotrajectory that follows the basin boundary towards the edge state, and each iteration also produces two trajectories that leave the boundary towards the on- and the off-state, respectively. These are the edge tracks analysed here. In addition, dedicated long simulations of the on-state, the off-state, and the edge state are available

at both $CO_2$ levels.

## Salinity readouts

We reduce the salinity output to two volume-weighted box means. The North Atlantic readout averages over 35°–80°N and the upper 13 ocean layers (down to 1025 m), and the Tropical Atlantic readout over 35°S–35°N and the upper 8 layers (down to 500 m). The edge-track output is already zonally averaged over the Atlantic. For the three-dimensional equilibrium output, we first apply the Atlantic basin mask of the model (with a southern border at 34°S) and take the zonal mean. Cells are weighted by cos(latitude) times layer thickness, and land cells are excluded.

At 285 ppm, we use 37 edge-track files and, at 360 ppm, 38 files, each containing one track towards each attractor. We classify the tracks by their final state rather than by their label in the edge-tracking output. The North Atlantic salinity averaged over the last 10% of each track is compared with its median over all tracks at that $CO_2$ level, and tracks above the median are assigned to the on-state. This disagrees with the original labels for two tracks per $CO_2$ level, none of which converges to its attractor according to the criterion below. From the edge-state equilibrium runs, the first 4 years (285 ppm) and 60 years (360 ppm) are discarded.

## Resilience measures

For each of the three states, we compute the mean $\mu$ and covariance matrix $C$ of the corresponding equilibrium run in the plane of the two salinity readouts. A point $x$ lies within the $4\sigma$ ellipse of a state if its Mahalanobis distance $\sqrt{(x-\mu)^\top C^{-1}(x-\mu)}$ is at most 4. The convergence time of a track is the time between its last visit to the edge-state ellipse and its first subsequent entry into the ellipse of its target attractor. Tracks that never enter the edge ellipse, or never reach the target ellipse afterwards, are counted as not converged. Many tracks at 285 ppm are short, and only 4 of 37 tracks converge to each of the on- and off-states. At 360 ppm, 19 of 38 tracks converge to the on-state and 33 of 38 to the off-state.

Table S3: Resilience quantities for PlaSim-LSG in the deep-box salinity plane.

| $CO_2$ | State | Mean conv. time (yr) | Edge distance ( psu) | Local resilience | AMOC (Sv) |
|---|---|---|---|---|---|
| 285 ppm | on | 105.3 | 0.050 | 0.0357 | 14.7 |
| 285 ppm | off | 116.0 | 0.204 | 0.0115 | 1.1 |
| 360 ppm | on | 150.6 | 0.110 | 0.0368 | 15.0 |
| 360 ppm | off | 74.8 | 0.093 | 0.0087 | 1.7 |

The distance between the edge state and an attractor is the gap between their $4\sigma$ ellipses along the line connecting the two centres, $\max(0, |\mu_{\text{att}} - \mu_{\text{edge}}| - r_{\text{edge}} - r_{\text{att}})$, where $r$ is the $4\sigma$ radius of each ellipse in that direction. For the on-state, this is the minimal-critical-shock proxy of the main text. The local-resilience proxy is $[2000\sqrt{\lambda_{\max}(C)}]^{-1}$, where $\lambda_{\max}(C)$ is the largest eigenvalue of the on-state covariance. The quantity $2\sqrt{\lambda_{\max}(C)}$ is the long axis of the $1\sigma$ ellipse in psu, and the additional factor 1000 expresses it in units of $10^{-3}$ psu. This proxy reflects the size of the fluctuations around the on-state and is not an eigenvalue of the underlying dynamics. Basin volume is not estimated, since the edge tracks start close to the basin boundary and do not sample a reference region of state space. Table S3 lists all resulting values.

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